Robust CSI-Based Wireless Channel Condition Classification With Cross-Environment Generalization

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Abstract

Wireless sensing based on Channel State Information (CSI) is a promising technique for indoor environment analysis and wireless channel condition classification However, current machine learning models sometimes fail to generalize well across diverse environments because of variations in signal propagation and environmental conditions. This paper proposes a robust wireless channel condition classification system based on CSI that is able to discriminate between Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) conditions with enhanced cross-environment generalization. The proposed system uses data preprocessing, normalization, feature extraction and sequence generation techniques on the CSI amplitude data to improve the data quality and the model performance. We implement and comparatively evaluate several machine learning (ML) and deep learning (DL) architectures, namely Support Vector Machine (SVM), Random Forest (RF), CNN-LSTM, and CNN-Transformer, for LOS/NLOS classification in unseen wireless environments. The experimental results show that deep learning models, especially CNN-LSTM achieved 91% and CNN-Transformer achieved 99% classification accuracy on unseen environments, outperforming traditional machine learning approaches on raw CSI data. The CNN-LSTM model achieved an AUC of 0.99, and the CNN-Transformer model provided effective attention-based feature learning with an AUC of 0.94. The results show that the appropriate preprocessing and hybrid deep-learning methods can greatly improve the accuracy and robustness of wireless channel condition classification for practical smart indoor applications.

Keywords: Channel State Information (CSI), WiFi Sensing, LOS/NLOS Classification, Cross-Environment Generalization, CNN-LSTM, CNN-Transformer

1. Introduction

Recently, wireless systems are evolving beyond their traditional role of data transmission due to the rapid growth of wireless communication, the Internet of Things (IoT), and smart technologies. Modern wireless networks have been widely used for intelligent sensing applications such as recognition of human activity, indoor positioning, healthcare monitoring, smart homes, and industrial [1], [2], [3], [4]. Recent advances in artificial intelligence and deep learning have further enhanced the abilities of wireless systems to interpret and understand changes in their surrounding environment [2], [4], [5]. Various wireless sensing approaches have been studied, among which WiFi sensing has received much attention recently due to its ability to leverage existing WiFi infrastructure, without requiring additional sensing devices [1], [2], [4]. CSI is one of the most valuable information sources in WiFi sensing and describes how the wireless signals are modified during propagation between a transmitter and receiver [3], [4], [6]. CSI is a representation of the changes due to reflection, obstruction and environmental change and can be used to identify different wireless channel conditions such as LOS and NLOS [3], [6], [7]. Correct classification of these conditions is crucial to improving communication reliability, localization systems, adaptive wireless networks and future intelligent communication technologies [7]. However, the development of CSI-based models that can work reliably in different environments is still a challenging research problem [5], [6], [8].

Several technologies have been investigated for wireless environment sensing and LOS/NLOS classification. Radar based sensing systems use the reflection of electromagnetic waves to sense objects and estimate the propagation conditions with high accuracy. But these systems have dedicated hardware and specialized signal-processing infrastructure, higher deployment costs and higher power consumption [4]. Vision-based approaches, based on the use of cameras and computer vision techniques, can provide detailed information about the environment but are limited by privacy issues, light dependency and occlusion problems [2], [4]. Other techniques based on sensors such as infrared, ultrasonic, LiDAR and wearable sensors have also been used for environment monitoring [9]. These techniques can sense effectively, but generally require additional infrastructure and maintenance. In contrast, WiFi sensing is a practical and low-cost alternative that leverages existing wire-

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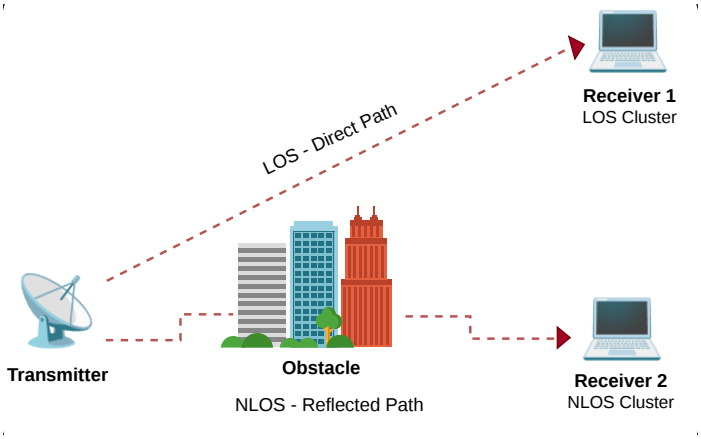


Figure 1: LOS vs NLOS Diagram

less communication systems. Unlike traditional signal measurements like Received Signal Strength Indicator (RSSI), CSI captures fine-grained variations in signal amplitude and phase and gives detailed information on each WiFi [10], [11]. Thus, CSI is more suitable for understanding complex wireless propagation characteristics and developing reliable LOS/NLOS classification systems.

Recent works have shown promising results on wireless channel classification using WiFi CSI with machine learning and deep learning techniques. Statistical features extracted from CSI signals have been widely used in traditional ML methods like SVM, RF, and other classification algorithms [2]. Recently, deep learning models such as CNN, LSTM and Transformer-based architecture have become popular due to their ability to learn complex pattern automatically from raw CSI measurements [10], [12]. However, despite these improvements, many existing approaches are evaluated on randomly divided training and testing data collected from the same environment. Such evaluation metrics are often not representative of real-world deployment scenarios where a model trained in one environment may need to work in a completely different environment. The change of the room layout, the existence of obstacles, the position of devices and the path of signal propagation may have a great impact on the CSI pattern, which will reduce the model performance [13], [14]. Some existing works have tried to solve this problem by transfer learning, domain adaptation and data augmentation techniques. However, these methods often require additional data from the target environment or complex re-training procedures. Robust CSI-based classification that can generalize to different environments without further adaptation is still an important research challenge [13], [15], [16].

In this work, we propose a robust CSI-based wireless channel condition classification framework with cross-environment generalization to overcome the limitations of existing works. The goal of this paper is to design and evaluate models capable of identifying LOS and NLOS scenarios, yet sustaining performance in various environments. The main contributions are:

- We developed a preprocessing pipeline to process raw CSI measurements and extract CSI amplitudes and representative features for LOS/NLOS classification.
- To effectively learn spatial and temporal characteristics from sequential CSI amplitude data for robust LOS/NLOS classification, two deep learning architectures (CNN-LSTM and CNN-Transformer) were designed and implemented.
- To assess their generalization capability under propagation conditions that were not previously seen, the developed models were tested on unseen environments.

The rest of this paper is organized as follows: Section 2 presents a literature review on existing CSI-based wireless sensing systems; related research works and research gaps. In section 3, the proposed system model is described along with dataset description, schematic model architecture, adopted methodologies and system requirements. The experimentation and model evaluation is presented in Section 4 which includes experimental setup, results, validation and performance analysis. Section 5 concludes and gives directions for future research.

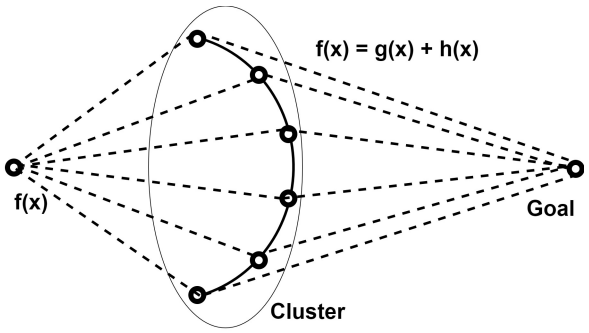


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2. Related Work

Wi-Fi sensing has slowly developed from traditional signal-based methods to directly using CSI. CSI describes how a wireless signal is affected while transmitting from the transmitter to receiver and also provides information for different subcarriers of Wi-Fi signal. Early work by Li et al. [17] used statistical features extracted from CSI together with SVM to identify LOS and NLOS conditions for indoor Wi-Fi localization. These statistical features, which are measures such as kurtosis, mean, skewness, standard deviation and rician-k factor, were calculated manually from CSI. Their method achieved an NLOS identification rate of 94.12% and improved over earlier threshold-based approaches, but they depended on pre-defined features and the authors noted that fixed thresholds could be strongly affected by changes in the environment and wireless devices. Choi et al. [18] later used an RNN to learn the time varying pattern of CSI for NLOS identification. Their work addressed difficulty of identifying an LOS path from a signal

CSI measurement by using a sequence of CSI measurements instead. Nguyen et al. [19] further proposed a CNN-LSTM model, where the CNN learned patterns across CSI subcarriers and the LSTM learned how these patterns changed over time. This combination improved the classification performance compared with an LSTM-based approach alone. However, these studies were mainly evaluated in indoor WLAN environments, and their ability to work in different or previously unseen environments was not fully investigated.

In further research, it has been identified that Wi-Fi sensing conditions and methods could handle changes in the sensing environment. Damodaran et al. [20] evaluated Wi-Fi CSI-based human activity recognition under both LOS and NLOS conditions. It showed that CSI can be used to recognize activities such as walking, sitting, standing, and running even when the direct signal path is blocked. Their study demonstrated that changes in CSI caused by human movement can still provide useful information for sensing in NLOS environments. However, changes in the surrounding environment can change the CSI distribution and reduce the performance of a model trained in different environment. Yang et al. [21], for example, studied CSI-based indoor localization in changing environments and used domain adaptation, which modifies a trained model using some data from a new environment so that it can work better in that environment. Their approach improved localization performance because it required only a small amount of newly collected data from the new environment. Although this reduced the need to build the complete model again, it still required data from the new environment. AirFi, proposed by Wang et al. [13], took a different approach by using domain generalization. In this the model is trained using multiple source environments and learns features that are less dependent on one particular environment. This allowed the model to recognize gestures in an unseen environment without collecting CSI data from that new environment. However, the work focused on gesture recognition rather than direct LOS/NLOS classification, and the problem of large differences between environments remained challenging.

The earlier limitations in Wi-Fi sensing studies were also examined more systematically as newer Wi-Fi technologies became available. Cominelli et al. [22] investigated CSI-based sensing using Wi-Fi 6 and considered multiple people, environments, and activities. Multiple-Input Multiple-Output (MIMO) uses multiple transmitting and receiving antennas. It provides CSI information from different paths travelled by the signal between the transmitter and the receiver in the environment. The study found that MIMO and higher spectral resolution, which provides finer information across the frequency subcarriers, can be more useful for sensing than simply increasing the signal bandwidth. However, the study also showed that achieving an environment-independent Wi-Fi sensing system remains difficult. More recently, Strohmayer and Kampel [23] studied data augmentation, which means creating modified versions of existing CSI samples so that a model is exposed to a wider range of signal variations during training. Their work specifically examined cross-domain recognition between LOS and NLOS through-wall scenarios and between different antenna systems. The results showed that suitable combinations of augmentation methods can improve cross-scenario and cross-system performance, but the benefit depended on the particular domain change. Thus, although domain generalization and augmentation improved robustness, they did not completely remove the effect of environmental and hardware differences.

Recent research has therefore moved further toward understanding and improving cross-domain generalization. In this, a model is trained under one set of conditions is expected to work under different conditions. Brunello et al. [24] showed that the problem is not limited to changes between different environments or devices; CSI can also change over time because of multipath effects and changes in surrounding objects. Their study showed that the performance of CSI-based sensing can therefore decrease when a model is used over longer periods, highlighting the importance of temporal generalization. More recently, Jiang et al. [25] investigated foundation-model-based learning for zero-shot cross-domain Wi-Fi sensing using more than 1.3 million CSI samples from 14 datasets collected using different devices, frequency bands, and bandwidths. Their results showed that increasing the amount and diversity of training data can improve performance on unseen domains, although the authors found that data diversity is still a major limitation and simply increasing model size provides only limited improvement. Despite these developments, robust LOS/NLOS classification across previously unseen environments remains insufficiently explored. Most recent cross-domain studies focus on human activity recognition, gesture recognition, or localization rather than directly classifying the wireless channel as LOS or NLOS. In addition, some methods require data from the target environment for adaptation, while others depend on large and diverse datasets. Therefore, there is still a need for methods that can learn useful representations directly from sequential CSI data and evaluate LOS/NLOS classification using a strict environment-wise split, where the testing environments are not used during training. This motivates the present work, which investigates CNN-LSTM and CNN-Transformer models for sequential CSI-based LOS/NLOS classification and evaluates their ability to generalize to previously unseen environments.

3. Proposed System/Model

In this section, we introduce the proposed CSI-based wireless channel condition classification framework, including the description of the dataset, system architecture, methodologies adopted, and the algorithms implemented for LOS/NLOS classification with cross-environment generalization.

3.1. Dataset Description:

In this study, the experiments were carried out based on the publicly available Massive MIMO Multi-User Channel Correlation Dataset proposed by Du et al. [26]. The dataset is collected using the Argos massive MIMO platform at Rice University and includes real over-the-air (OTA) wireless channel measurements in near-stable outdoor environments. The measure-

Table 1: Comparison of various methods and their contributions and limitations.

Reference	Primary Focus	Architecture/ Methodology	Key Contribution	Limitations
Li et al. [17] (2017)	Direct LOS/NLOS identification & NLOS error mitigation.	SVM for LOS/ NLOS identification; regression model for NLOS mitigation.	Shows that CSI can provide useful information for NLOS identification; Combined multiple CSI features instead of relying on a single threshold	Relies on manually extracted features; mainly indoor scenarios; performance can depend on the environment and device
Choi et al. [18] (2018)	Direct LOS/ NLOS identification using temporal CSI	RNN using sequences of CSI measurements.	Addressed the difficulty of detecting LOS path from a single low resolution CSI measurement by learning CSI variation over time.	Commodity WLAN devices have limited resolution; evaluated in specific environments; cross-env generalization not main focus.
Nguyen et al. [19] (2018)	Direct LOS/NLOS identification	Joint CNN-LSTM/RNN	Combined frequency/ subcarrier information with temporal information; reported improved performance compared with LSTM alone.	Mainly evaluated in indoor WLAN environments; unseen-environment performance was not extensively studied.
Damodaran et al. [20] (2020)	Human Activity Recognition (HAR) and fall detection under LOS and NLOS	SVM with waveletbased preprocessing and feature extraction; LSTM operating on denoised CSI.	Demonstrated that CSI can support activity recognition even when the direct path is blocked; evaluated activities such as running, walking, sitting, and standing in both LOS and NLOS settings.	Focus is HAR rather than direct LOS/NLOS classification; experiments are limited to indoor settings environment transfer/generalization remains an issue.
Yang et al. [21] (2021)	CSI based indoor localization under changing environments	Transfer learning; Integration of Global and Local Metrics for Domain Adaption (IGLDA)	Addressed the problem of CSI changing when the environment changes; improved reported localization accuracy from 26.3% to 82.2% while recollecting only 37.5% of the data	Requires data from the new environment; designed for localization rather than LOS/NLOS classification; does not provide completely unseen-environment operation.
Wang et al. [13] (2022/ 2024)	Wi-Fi CSI gesture recognition in unseen environments	Domain generalization; uses feature augmentation	Moves beyond domain adaptation by attempting to work in an unseen environment without collecting CSI from that environment; improves robustness across environments	Focuses on gesture recognition rather than LOS/NLOS classification; still depends on multiple source environments; large domain differences can remain difficult.
Cominelli et al. [22] (2023)	Understanding CSI sensing capabilities and limitations	Wi-Fi 6 CSI; experiments across people, environments and activities	Systematically studied the effect of modern WiFi features; showed that MIMO and higher spectral resolution can be more beneficial than simply increasing bandwidth	Not a new LOS/NLOS classifier; focuses mainly on HAR; study shows that truly environmentindependent CSI sensing is still difficult.
Strohmayer & Kampel [23] (2024)	Cross-domain CSI generalization, including LOS/NLOS for HAR	CSI amplitude spectrograms; EfficientNetV2; several data augmentation techniques	Investigated if augmentation can improve generalization b/w different scenarios and h/w; showed that some combinations of simple augmentation methods improve cross-domain performance	Focuses on HAR rather than direct LOS/NLOS classification; augmentation effectiveness depends on the domain shift; no single augmentation method works for all cases .
Brunello et al. [24] (2025)	Temporal generalization and stability of WiFi CSI sensing	Methods for Indoor Positioning (IP) and HAR; how CSI changes over time	IP tasks depending on static reflections were sensitive to temporal changes, HAR based on dynamic reflections more robust.	Focuses on temporal generalization, not direct LOS/NLOS classification. It primarily evaluates indoor positioning and HAR.
Jiang et al. [25] (2025)	Zero-shot cross-domain Wi-Fi sensing	Masked Autoencoding style foundation-model pretraining on large heterogeneous CSI data	Used >1.3 million CSI samples from 14 datasets and 4 devices; showed that increasing data scale and diversity improves unseen-domain performance	Large and diverse CSI datasets are still limited; focus is on HAR, gesture recognition and user identification rather than direct LOS/NLOS classification.

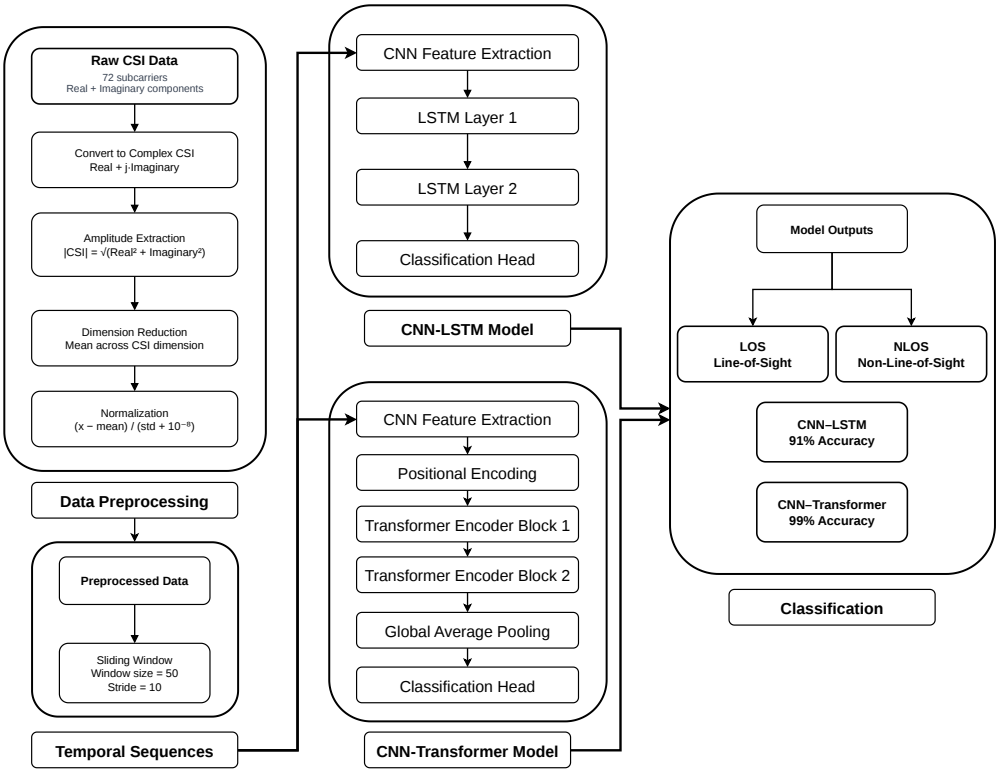


Figure 3: Flowchart

ment system is a 64-antenna (8 x 8) massive MIMO base station built from WARP Software Defined Radio (SDR) modules in the 2.4 GHz WiFi ISM band with 20 MHz bandwidth. During data acquisition the mobile users transmitted IEEE 802.11 training frames and the base station recorded the received IQ samples and estimated the wireless channel based on pilot-based channel estimation. The measurement process was performed with a dedicated calibration node to compensate the antenna phase mismatch of different RF chains, enabling phase-consistent CSI measurements.

The dataset contains wireless channel measurements from four LOS and five NLOS clusters. Each cluster is of the order of 4 m in diameter and holds measurements of more than 25 uniformly distributed user locations. This results in more than 225 measurement locations and more than 11,000 unique channel vectors. At each location, we collected CSI measurements over 140+ frames for the 52 Orthogonal Frequency Division Multiplexing (OFDM) data subcarriers, to capture the variations of the wireless propagation channel. In this work, the raw complex CSI measurements are pre-processed by extracting the CSI amplitude values and building sequential CSI representations. These will then be used to train and evaluate machine learning and deep learning models for LOS/NLOS classification with cross-environment generalization.

The original dataset paper reports measurements for the 52 data-carrying OFDM subcarriers. However, the released raw CSI traces used in this work contain CSI measurements for 72 subcarriers. Hence, in the pre-processing stage, raw CSI amplitudes of all 72 subcarriers were extracted and used to maintain the complete channel information.

The raw WiFi CSI dataset is different from traditional machine learning datasets with manually engineered features, as it does not have pre-defined features. Instead, each sample is a set of CSI measurements over multiple OFDM subcarriers. Each subcarrier corresponds to the wireless channel response at a specific frequency. These subcarriers have detailed information about the signal propagation effects such as attenuation, multipath propagation, reflection, diffraction, scattering and environmental changes. As a result, the raw dataset contains the consecutive CSI measurements on subcarriers rather than explicit statistical features. This enables the feature extraction for deep learning models to learn discriminative representations directly from the underlying wireless channel characteristics.

The raw CSI traces were saved in a four-dimensional tensor (frames, 72, 1120, 2), where frames are sequential CSI measurements, 72 is the number of OFDM subcarriers, 1120 is the pilot measurement points, and 2 is the real and imaginary parts of the complex CSI values.

3.2. Schematic Layout of the Proposed System

The proposed system is based on a CSI processing and classification pipeline for identifying the LOS/NLOS wireless channel condition. The general work-flow includes CSI frame input,

Table 2: Number of samples in dataset

File	Samples
LOS Cluster_1	8140
LOS Cluster_2	8159
LOS Cluster_3	8112
LOS Cluster_4	8101
NLOS Cluster_1	6081
NLOS Cluster_2	4054
NLOS Cluster_3	4058
NLOS Cluster_4	4057
NLOS Cluster_5	4063

Table 3: Number of samples in each file

Class	Labels	Samples
NLOS	0	22313
LOS	1	32512
Total	-	54825

preprocessing, temporal sequence generation, ML and DL classification, result analysis and final output generation. This flow has been illustrated in Figure 3.

At first raw CSI frames are collected from the wireless dataset. They are checked for missing values, infinite values and invalid measurements before pre-processing. No major data quality issues found, which meant that the dataset is already clean and ready for further processing. The pre-processing step consists of reconstruction of complex-valued channel information by combining the given real and imaginary parts, amplitude extraction, dimensionality reduction, normalization. This pre-processed data is used to train ML models. Also, temporal sequences are generated for the DL models by a sliding window mechanism with overlapping CSI sequences and window size of 50 and stride of 10. The generated sequences are then fed to multiple DL classification models such as CNN-LSTM and CNN-Transformer architecture. Past tentances issue

The CNN layers extract spatial CSI features and LSTM and Transformer blocks capture temporal dependencies and long-range sequential relationships inside wireless signals. The system also offers cross- environment testing where models are trained on certain selected environments and tested on unseen environments to analyse robustness and generalization capability.

3.3. Methodologies Used

The proposed methodology presents a deep learning-based framework for LOS and NLOS classification using WiFi CSI. The overall workflow consists of CSI data pre-processing, sequence formation, model development, training, and performance evaluation on unseen data. The raw CSI measurements are first preprocessed to obtain the amplitude information from the complex CSI values, which then undergoes dimensionality reduction and normalization. The pre-processed data is then divided into two parts – training and testing set to prevent data leakage and evaluate the model performance strictly on unseen testing data. This training set of pre-processed data is then used to train ML models such as SVM and RF. The pre-processed data is used for temporal sequence generation which is required for the DL model training. The temporal sequences are generated by using sliding window size of 50 and stride 10, and then the dataset is divided into two parts – training set and testing set. There are two DL models used separately, i.e. CNN-LSTM and CNN-Transformer.

The CNN-LSTM model consists of CNN Feature Extraction Block, LSTM Layer 1, LSTM Layer 2, Classification Head as shown in the flow chart diagram 2. Similarly, the CNN-Transformer model consists of CNN Feature Extraction Block, Positional Encoding, Transformer Encoder 1, Transformer Encoder 2, Global Average Pooling, Classification Head as shown in the flow chart Figure 3.

In Result and Discussion part: For model evaluation we have used accuracy, precision, recall, f1-score, confusion matrix and roc curve. The complete methodology provides an end-to-end pipeline for the pre-processing and classification of WiFi CSI data into LOS and NLOS and enables a comparative assessment of CNN-LSTM and CNN-Transformer model.

3.3.1. Data Pre-processing:

We have 4 different components in raw csi dataset, which are FrameCompleteTime, RSSI, Pilot_Samples and mob_tx_gain. We use the Pilot_Sample for our project. It is initially stored as 4-Dimensional array, one example is (8140,72,1120,2). Upon further inspection, we can see that when we expand dimension the actual data is stored in 6-Dimensional array, example (8140,1,1,72,1120,2). Now we are able to match it with the terminology given in the github repository of the original dataset. It specifies that the dimensions are total frames, cell, user equipment, antennas, individual time samples, In-phase and Quadrant path (IQ path). IQ path is represented as In-phase component which is the real part of complex csi value and Quadrant component which is the imaginary part of the complex value csi. But we only need the 4-Dimensional representation. So, we represent the data as:

$$X \in \mathbb{R}^{n \times m \times t \times q}$$

where: n = number of frames, m = number of antenna, t = number of time samples, q = real and imaginary components (m, t and q is same for each file in the dataset)

The raw CSI data were pre-processed in a memory-efficient, chunk-based manner due to the large size of the dataset. We loaded each chunk of the data sequentially, and combined the real and imaginary parts of CSI to reconstruct the complex-valued channel response.

$$Z_{n,m,t} = X_{n,m,t,I} + jX_{n,m,t,Q} \tag{1}$$

such that:

$$Z \in \mathbb{C}^{n \times m \times t}$$

The CSI amplitude was then calculated as the magnitude of the complex values using the following formula:

$$A = \sqrt{X_{n,m,t,I}^2 + X_{n,m,t,Q}^2} \tag{2}$$

Since each antenna contained 1120 time-domain I/Q samples for each recorded frame, the magnitude values were averaged over the time-sample dimension to obtain a compact 72-dimensional representation while retaining the antenna-wise channel characteristics. Dimensionality Reduction mathematical representation:

$$F_{n,m} = \frac{1}{t} - \sum_{i=1}^{t} A_{n,m,t} \tag{3}$$

such that:

$$F \in \mathbb{R}^{n \times m}$$

All chunks were processed and concatenated to form the complete dataset. Finally, z-score normalization was performed by subtracting the global mean and dividing by the standard deviation, resulting in standardized amplitude data that could be used for subsequent machine learning and deep learning model training.

$$\hat{F}_{n,m} = \frac{F_{n,m} - \mu_F}{\sigma_F + 10^{-8}} \tag{4}$$

where:

$$\mu_F = \text{mean}(F), \quad \sigma_F = \text{std}(F).$$

3.3.2. Temporal Sequence Generation:

There are 9 clusters/environments in total present in our raw CSI dataset. Out of 9 clusters, 4 clusters are LOS and 5 clusters are NLOS. While pre-processing the dataset we made sure to maintain the clusters and prevent shuffling them. After pre-processing the dataset, we divided it cluster-wise into training set (2 LOS and 3 NLOS clusters) and testing set (2 LOS and 2 NLOS clusters). We used the same division criteria for all the models, so that the results are comparable. The Temporal Sequence Generation was done on training set and testing set individually to prevent the data samples from mixing and causing a data leakage. Because the sliding window size for temporal sequence generation is 50 and the stride is 10, some of the data samples were omitted in both the training and testing set files. In total, 2416 test sequences and 3027 train sequences were generated, which can be calculated as follows:

$$\text{Testing/Training Sequences} = \sum_{i=1}^{n} \left(\left\lfloor \frac{N_i - 50}{10} \right\rfloor + 1 \right) \tag{5}$$

where: n = number of files in testing/training set, N_i = number of samples in i^{th} testing/training file

3.3.3. Machine Learning Methods:

The pre-processed data was used to train the ML models. The methods used are SVM which aims to find the best hyperplane that can separate the data points of different classes with the maximum margin [27] and RF which uses an ensemble of a large number of decision trees generated from random subsets of training data and features with the majority vote rule for classification [28]. These models were employed to learn discriminative patterns between LOS and NLOS environments using amplitude-based CSI features.

- SVM: The processed CSI amplitude data set was divided into training and testing data sets by a cluster-wise split. For the testing set, unseen data, i.e. the samples used for evaluation were completely removed from the training process. This allows for a realistic measure of the model's ability to generalize to previously unseen channel conditions, and hence implements cross-environment evaluation, as the model is tested on new data. The clusters that were selected were merged to obtain the final datasets. A custom data loading function was employed to load the .npy files and to label the data in binary (LOS = 1, NLOS = 0).We trained an SVM classifier with a Radial Basis Function (RBF) kernel, which is used for non-linear dataset, on the training data with probability estimation enabled. Then the trained model was used to make predictions on the testing samples and class probabilities. Finally, model performance was evaluated using classification accuracy, confusion matrix, precision, recall, F1-score, ROC-AUC and Precision–Recall (PR) curve to evaluate model LOS/NLOS classification performance. The decision boundary is determined by the Hyperplane Equation [27]:

  $$w^T x + b = 0 \tag{6}$$

  Where: w = weight vector, x = input feature vector, b = bias term

  The geometric distance between the support vectors is maximized using the Margin Formula [27]:

  $$Margin = \frac{2}{\|w\|} \tag{7}$$

  Where: w = weight vector

- RF: The Random Forest model was implemented by loading the processed CSI amplitude data from the predefined training and testing cluster files and assigning binary labels for the LOS and NLOS samples. The final training and testing datasets were constructed by combining samples from all files. Then, a Random Forest classifier with 100 decision trees was trained on the training data to learn the difference between LOS and NLOS channels. The trained model was tested on the unseen testing data and its performance was evaluated using classification accuracy along with a confusion matrix to measure the effectiveness of the classifier. The splitting quality at each node within the decision trees is evaluated using the Gini Impurity criterion [28]:

  $$G = 1 - \sum_{i=1}^{n} p_i^2 \tag{8}$$

  Where: p_i = probability of class, n = number of classes

3.3.4. Deep Learning Methods:

Deep learning models were trained using sequential CSI amplitude data, obtained from the pre-processed raw CSI traces. The CNN-LSTM model is a combination of the extraction of convolutional features and the learning of temporal features based on LSTM. The CNN-Transformer model is a combination of the extraction of convolutional features and the learning of long-range dependencies in sequences based on the Transformer model which is used for LOS/NLOS classification.

- CNN – LSTM: The preprocessed CSI amplitude files were labeled as LOS (1) or NLOS (0) and then data was split environment-wise to prevent data leakage, by using five environments (two LOS and three NLOS clusters) for training and four unseen environments (two LOS and two NLOS clusters) for testing. To capture temporal dependencies, we employed a sliding-window approach to produce sequential CSI samples of size 50×72 with a window size of 50 and a stride of 10, and then aggregated them into training (3027 sequences) and testing (2416 sequences) datasets. A hybrid CNN-LSTM model was designed with two Conv1D layers with 64 and 128 filters, each followed by Batch Normalization and Max Pooling, to extract spatial features and reduce dimensionality, followed by two stacked LSTM layers with 128 and 64 units, to learn temporal relationships. A fully connected layer with L2 regularization and Dropout was used prior to the final sigmoid output layer for binary LOS/NLOS classification. The model was trained using Adam optimizer (learning rate = 0.001), binary cross entropy loss and the training data was split into training and validation data using 80:20 stratified split. Training was performed for up to 50 epochs with a batch size of 32, and Early Stopping was employed to monitor the validation loss and restore the best-performing model weights, thereby reducing overfitting. Finally, the trained model was evaluated on the unseen testing environments, where sigmoid outputs were thresholded at 0.5 to obtain class predictions, and performance was assessed using accuracy, precision, recall, F1-score, confusion matrix, and ROC analysis.

  Convolution Operation:

  $$S(i, j) = (X * K)(i, j) \tag{9}$$

  Where: X = input feature map, K = kernel/filter, S = output feature map

  Forget Gate Equation:

  $$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{10}$$

  Where: f_t = forget gate, σ = sigmoid activation function, W_f = forget gate weight, h_{t-1} = previous hidden state value, x_t = input value, b_f = forget gate bias

  Input Gate Equation:

  $$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{11}$$

  Where: i_t = input gate, σ = sigmoid activation function, W_i = input gate weight, h_{t-1} = previous hidden state value, x_t = input value, b_i = input gate bias

  Candidate Memory Equation:

  $$\tilde{C}_t = tanh(W_c[h_{t-1}, x_t] + b_c) \tag{12}$$

  Where: $\tilde{C}_t$ = candidate memory, tanh = hyperbolic tangent activation function, W_c = candidate memory weight, h_{t-1} = previous hidden state value, x_t = input value, b_c = candidate memory bias

  Output Gate Equation:

  $$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{13}$$

  Where: o_t = output gate, σ = sigmoid activation function, W_o = output gate weight, h_{t-1} = previous hidden state value, x_t = input value, b_o = output gate bias

- CNN-Transformer: The CNN-Transformer model was implemented using sequential CSI data generated through a sliding window approach, where each input sample consisted of a sequence of 50 consecutive CSI observations with 72 subcarrier amplitude features. The architecture combined convolutional neural networks with a Transformer encoder to effectively learn both local and global feature representations. Initially, two one-dimensional convolutional layers, each followed by batch normalization, max-pooling, and dropout, were employed to extract discriminative local spatial features, reduce noise, and decrease the sequence length, thereby lowering computational complexity. Positional encoding was subsequently applied to preserve the temporal order of the extracted feature maps before they were processed by two stacked Transformer encoder blocks. Each encoder includes multi-head self-attention, residual connections, layer normalization, feed-forward networks, and dropout to learn long-range temporal dependencies in CSI sequences. The resulting feature representations were combined with a global average pooling layer and forwarded to fully connected layers with dropout regularization, leading to a sigmoid output layer for binary LOS/NLOS classification. The model was implemented using Keras Functional API and was trained using Adam optimizer, binary cross-entropy loss and accuracy as the evaluation metric. To avoid overfitting and to recover the model weights with the best performance, early stopping was done based on validation loss. Finally, the trained model was tested on an unseen testing dataset, where the probability scores were converted into binary class labels using a threshold of 0.5. The model performance was evaluated using accuracy, confusion matrix, ROC curve and classification metrics such as precision, recall and F1-score.

  Self-Attention Mechanism:

  $$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{14}$$

Where: Q = Query matrix, K = Key matrix, V = Value matrix, d_k= Dimension of key vectors, softmax = Normalization function used to convert scores into probabilities

Softmax Equation:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_i}} \tag{15}$$

Where: p(y_i)= Probability of class i, z_i= Output score for class i, e = Exponential function, n = Total number of classes

3.4. System Requirements

Implementation was done using Python as the main programming language in Google Colab and Jupiter Notebook environments that provided GPU-enabled execution support for deep learning model training. The main libraries and frameworks used are: NumPy for numerical computations, Pandas for data manipulation, Matplotlib and Seaborn for data visualization, h5py for loading raw CSI traces in HDF5 format, Scikit-learn for implementing SVM and RF models, and TensorFlow/Keras for implementing CNN-LSTM and CNN-Transformer architectures.

Algorithm 1: AnchorZoneDiscovery(azquery)

1 [1] **if** *(azquery.flag == 0)* **then**

2 Flag 0 means, previous sender was anchor node azquery.sender = selfAddress transmit azquery to one-hop neighbors start next hop discovery using A3DR protocol with Equation (**??**) (azquery.flag == 1) Flag 1 means, previous sender was one-hop neighbor of anchor node

3 azquery.sender = selfAddress azquery.senderpos = selfposition azquery.flag = 2 respond azquery to previous sender according to A3DR protocol using Equation (**??**)

4. Experimentation and Model Evaluation

In this section, the experimental setup, results, validation and comparative analysis of the implemented ML and DL models for LOS/NLOS classification with CSI Amplitude data are described. The models were evaluated on unseen testing environments to analyze classification capability, robustness and generalization performance.

4.1. Experimental Setup

A table needed We evaluate the proposed models on a publicly available Massive MIMO Multi-User Channel Correlation dataset collected using the Argos platform at the Rice University campus. Measurements were taken in outdoor near stable environments where a massive MIMO base station with 64 antennas (8×8) was communicating with mobile user nodes in the 2.4 GHz WiFi ISM band with a bandwidth of 20 MHz. The dataset contains nine measurement clusters, including four LOS clusters and five NLOS clusters. Each cluster covers an area of about 4 m in diameter and consists of measurements from more than 25 uniformly distributed user locations, thus, having more than 225 measurement locations. The LOS clusters represent locations with a direct line of sight between the base station and the mobile user, while the NLOS clusters contain locations where the direct line of sight is blocked by surrounding campus buildings, and hence the received signals propagate through reflections, diffraction and scattering. As a result, the multipath propagation characteristics and channel conditions are different across different clusters.

Since the dataset was collected in an outdoor campus environment, no indoor rooms nor hybrid indoor-outdoor scenarios are included in the dataset. Similarly, the exact distance of the transmitter to the receiver is not specified for each measurement location in the original dataset paper. However, all the user locations were distributed within the pre-defined LOS and NLOS clusters around the massive MIMO base station and a dedicated calibration node was placed approximately 20 m away from the base station to compensate the antenna phase mismatch during the channel acquisition. We exploit the diversity in the nine spatially separated clusters to evaluate the proposed models in different propagation environments and provide a realistic benchmark for assessing the cross-environment generalization.

The dataset consists of raw CSI traces with 72 subcarriers features. The dataset was split environment-wise into training set (5 files: 2 LOS and 3 NLOS) and testing set (4 files: 2 LOS and 2 NLOS). The testing set is referred as unseen data, i.e. the samples used for evaluation were completely removed from the training process, to prevent data leakage between environments. This allows for a realistic measure of the model's ability to generalize to previously unseen channel conditions, and hence implements cross-environment evaluation, as the model is tested on new data. The ML models were trained on the processed CSI data, while temporal sequences were generated using sliding window of size 50 and stride of 10 for DL models.

4.2. Results and discussions

To fully evaluate the proposed framework, we analyze its performance for LOS/NLOS channel condition classification and cross-environment generalization using both conventional machine learning and deep learning models. SVM and RF are chosen as the baseline machine learning models because of their successful performance in the CSI-based classification task. SVM is good at high dimensional data and finding the best decision boundary, while RF is good at noisy data and can learn complex non-linear relationships by ensemble learning. Two deep learning architectures, CNN-LSTM and CNN-Transformer, are used to further exploit the temporal characteristics embedded in sequential CSI data. The CNN-LSTM model marries convolutional feature extraction with temporal sequence learning, whereas the CNN-Transformer employs the self-attention mechanism to grasp long-range dependencies in CSI sequences. The performance of these models is evaluated in a sequence, starting from the analysis of SVM, followed by

RF, CNN-LSTM and CNN-Transformer. For each model, multiple performance metrics such as accuracy, precision, recall, F1-score, confusion matrix, Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC) are computed to provide a comprehensive evaluation of classification performance and to compare the ability of each approach in achieving robust LOS/NLOS classification in different environments.

SVM Results: The SVM classifier achieved an overall accuracy of 82.24% and a weighted F1-score of 0.83 which shows its ability to classify the LOS (1) and NLOS (0) channel conditions effectively. The confusion matrix shows that the NLOS (0) samples are correctly classified, but 4321 LOS (1) samples are misclassified as NLOS (0), reducing the recall of the LOS (1) class to 73%. From the probability distribution, it can be seen that most of the NLOS (0) samples are assigned probabilities close to 0, and most of the LOS (1) samples are assigned probabilities close to 1, resulting in an AUC of 0.9998. However, some LOS (1) samples are obtained below the default threshold of 0.5 of the probabilities, resulting in misclassification to NLOS (0). These samples probably correspond to the difficult propagation conditions in outdoor environments where multipath propagation, scattering, diffraction and channel fading cause the overlap of the CSI amplitude characteristics of LOS (1) and NLOS (0) channels. In general, the results indicate that the SVM exhibits excellent class separability but reduced sensitivity to detect some borderline LOS (1) samples.

Table 4: Classification Report of SVM

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0(NLOS)	0.65	1.00	0.79	8120
1(LOS)	1	0.73	0.85	16213
accuracy	-	-	0.82	24333
macro avg	0.83	0.87	0.82	24333
weighted avg	0.88	0.82	0.83	24333

Random Forest (RF) Results: The Random Forest model had a test accuracy of 53.82%, which means it had limited generalization to the unseen testing clusters. The training and testing sets were selected from different clusters of LOS and NLOS, which made the statistical properties of the amplitude of the CSI data different between the two sets. But, as Random Forest is based on hand-crafted feature distributions, it could not adapt to these unseen channel conditions well.

The confusion matrix shows that the model correctly classified 8,099 out of 8,120 NLOS (0) samples but only correctly recognized 4,998 out of 16,213 LOS (1) samples. The classifier was biased to predict the NLOS (0) class, as a significant number of LOS (1) samples (11,215) were misclassified as NLOS (0). This imbalance had a significant impact on the overall classification performance.

The classification report also confirms that the model is biased towards the NLOS class. The precision for the NLOS class was 0.42, which means that many samples predicted as NLOS were actually LOS, while the precision for the LOS class was 1.00, which indicates that the predictions for LOS were highly reliable when made. However, the model obtained a recall of 1.00 for the NLOS class but only 0.31 for the LOS class, meaning that most LOS samples were misclassified as NLOS. Therefore, the weighted F1-score of 0.51 reflects the overall imbalance in the classification performance of the model.

The ROC curve showed an AUC of 0.94 indicating good discrimination between the LOS and NLOS samples based on the predicted probabilities. However, the poor classification accuracy despite high AUC indicates that the default decision threshold had a strong bias towards the NLOS class, which resulted in poor overall classification performance on the unseen testing data.

CNN-LSTM Results: The CNN-LSTM model achieved a test accuracy of 96.52%, demonstrating its capability to model the spatial features of CSI subcarriers by the convolutional layers and the temporal dependencies of sequential CSI measurements by the stacked LSTM layers. The Classification Report shows strong and fair performance, where the LOS class achieves a precision of 1.00, a recall of 0.95 and an F1-score of 0.97, while the NLOS class obtained a precision of 0.91, a recall of 1.00 and an F1-score of 0.95. The weighted average precision, recall, and F1-score of 0.97 further validate the resilience of the model despite the little class imbalance in the test dataset.

The confusion matrix shows that all 803 NLOS (0) samples were correctly classified, and 1529 of the 1613 LOS (1) samples were correctly identified, with 84 LOS (1) samples incorrectly categorized as NLOS (0). This indicates that the model has a small tendency to misclassify ambiguous LOS (1) samples as NLOS (0), possibly due to channel fluctuations, noise, or multipath propagation that creates similarities between the CSI features of both classes. The few misclassifications however, show that the model is able to generalize well to unseen situations.

The ROC curve also had an AUC of 1.00 which demonstrates an excellent discrimination between the LOS and NLOS classes at different decision thresholds. The result indicates that CNN-LSTM is successfully learned very discriminant feature representations from the sequential CSI data, and the few classification errors are mainly due to the static decision threshold rather than the learned feature representations.

Table 5: Classification Report of CNN-LSTM

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0(NLOS)	0.79	1.00	0.88	803
1(LOS)	1.00	0.87	0.93	1613
accuracy	-	-	0.91	2416
macro avg	0.90	0.93	0.91	2416
weighted avg	0.93	0.91	0.92	2416

CNN-Transformer Results: The CNN-Transformer model

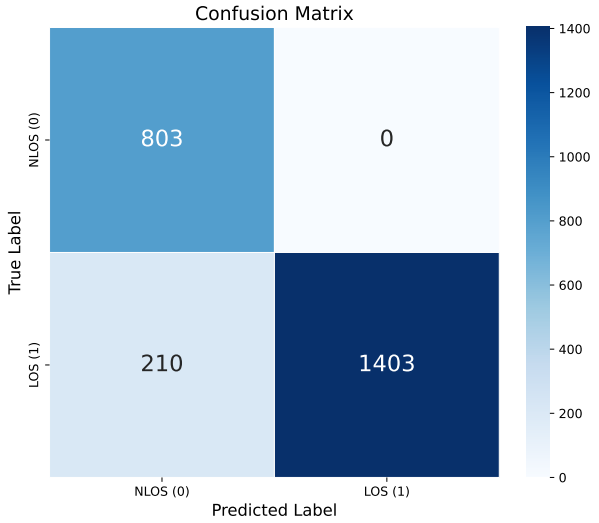


Figure 4: Confusion Matrix of CNN-LSTM

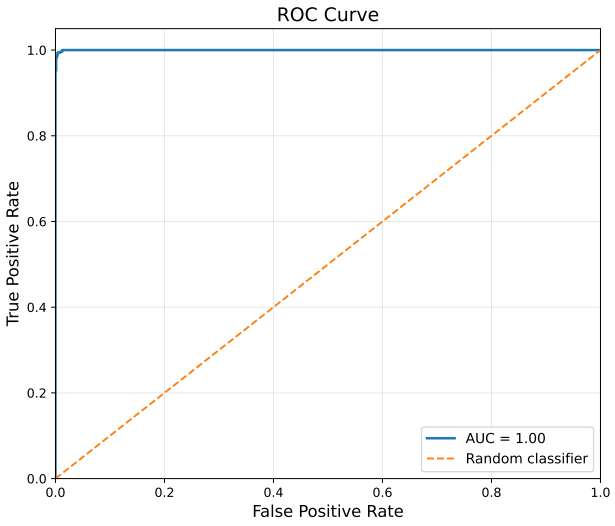


Figure 5: ROC Curve of CNN-LSTM

showed excellent performance for the LOS/NLOS classification with an overall testing accuracy of about 99.92%. The classification report returned precision, recall, and F1-score values rounded to 1.00 for both classes suggesting highly balanced classification performance across 803 NLOS (0) and 1613 LOS (1) testing sequences. The robustness of the proposed model was further confirmed through the confusion matrix, where it correctly classified 801 out of 803 NLOS (0) sequences and all 1613 LOS (1) sequences with only two NLOS (0) samples incorrectly predicted as LOS (1), with an extremely low misclassification rate. Similarly, the ROC curve was close to the upper left corner with an AUC of 1.0000, indicating good separability between LOS and NLOS classes for different decision thresholds. The improved performance is achieved by combining convolutional layers to extract discriminative local CSI features and Transformer encoder blocks to capture long-range temporal dependencies via multi-head self-attention. In addition, the positional encoding preserved the temporal order of the sequential CSI observations. Batch normalization, dropout and early stopping improved the model generalization and minimized the overfitting. The results show that the proposed CNN-Transformer architecture can make good use of the local spatial features and global temporal relations of sequential CSI data, which is reliable for robust wireless channel condition classification.

The experimental results verify that the proposed CNN-Transformer architecture achieves excellent performance for CSI-based LOS/NLOS classification with the overall testing accuracy of about 99.92%, where only two samples are misclassified among 2416 testing sequences. The convolutional feature extraction and Transformer-based self-attention are combined to effectively capture the local spatial features and long-range temporal dependencies in the CSI sequential data. The confusion matrix is almost perfect and the AUC value is very close to 1.0, which further proves the excellent discriminative ability and generalization ability of the proposed model, and it is very suitable for robust wireless channel condition classification.

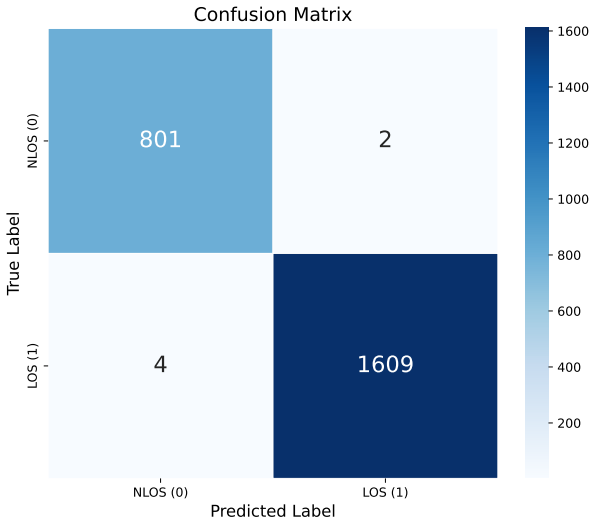


Figure 6: Confusion Matrix of CNN-Transformer

Table 6: Classification Report of CNN-Transformer

Class	Precision	Recall	F1-Score	Support
0(NLOS)	1.00	1.00	100	803
1(LOS)	1.00	1.00	1.00	1613
accuracy	-	-	1.00	2416
macro avg	1.00	1.00	1.00	2416
weighted avg	1.00	1.00	1.00	2416

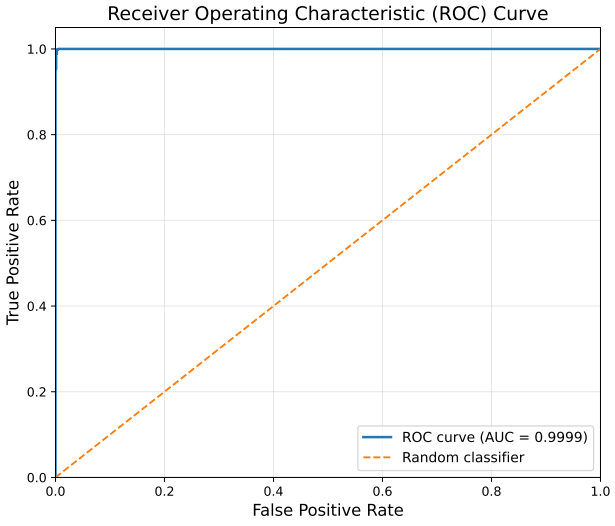


Figure 7: ROC Curve of CNN-Transformer

5. Conclusion

This work presents an energy-efficient anchor-based geographical routing protocol to detour the void with maximizing the lifetime of the anchor nodes as well as overall network. The proposed EAZR protocol includes an anchor zone for each side (i.e., left, right, top and bottom) of the network.

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